

Statistical Autofocus of Synthetic Aperture Sonar Images using Image Contrast Optimisation

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Abstract- In this paper, we present results of a technique for autofocus of stripmap synthetic aperture sonar (SAS) imagery. The technique of contrast optimisation is posed in a general Bayesian framework. The algorithm uses standard optimisation techniques on towfish motion parameters to minimise a cost function. The cost function consists of an image contrast measure, approximating the image likelihood, and a path likelihood measure, obtained from the towfish motion statistics. A number of simple image contrast measures are compared, and the optimal method for parameterising the towfish path is presented.

Results showing the cost-function surface for simulated and real data are given. These show the technique of contrast optimisation is viable, and demonstrate the need for some form of regularisation in the cost function, justifying the Bayesian approach. Results on simulated data, with known towfish motion, show a high degree of accuracy is possible when minimising this cost function, but can depend on the image quality measure used. Local gradient optimisation algorithms are shown to be inadequate for real data, due to the high level of noise on the contrast surface.

Preliminary results from a multiple-frequency interferometric SAS system, built by the Acoustics Research Group at the University of Canterbury are presented and ways to combine this extra information for autofocus purposes are discussed.

When combined with a bulk-error removal method, or INS measurements, this technique shows promise for high-quality autofocus of regions of interest.

I. INTRODUCTION

A sea-going towed synthetic aperture sonar (SAS) has the potential of producing high-resolution images of the seafloor. The largest problem is the blurring caused by unknown motion of the sonar towfish and turbulence within the imaging medium. Compensating for this unknown motion requires an accurate estimate of the variation of the towfish path from a straight line. Autofocus involves obtaining a path estimate from the collected data (as opposed to instrumentation) and compensating for this motion in the reconstructed image. Unlike Synthetic Aperture Radar (SAR) applications, current off-the-shelf inertial navigation systems (INS) are not accurate enough for this purpose; hence the need for autofocus.

Autofocus is usually approached in one of two different

This work was partly sponsored by the Department of the Navy, NAVSEA Coastal Systems Station, Panama City, Florida, as Grant N61331-00-1-G003.

S.A. Fortune was supported by a Bright Futures Doctoral scholarship from the Foundation for Research Science and Technology New Zealand.

ways [1,2,3], contrast optimisation, or phase gradient techniques. Contrast optimisation consists of changing various focus parameters of an image to maximise a contrast measure. In phase gradient techniques, the phase changes due to motion are estimated and used for focussing the image.

Section II gives some background on the SAS imaging system, and how known towfish motion is corrected in the images. Section III poses the image formation process as a general inverse problem. This problem is inherently ill-posed, as an infinite number of combinations of towfish motion and objects being imaged can produce the measured data. The problem of autofocus becomes a statistical one; that of choosing the most likely image and path combination. Sections IV and V describe the approach used to measure the most likely image and path, and the algorithm to find them. Standard optimisation techniques are used to perturb the path parameters, optimising this likelihood measure. The path is parameterised so that an arbitrary path can be represented up to the desired precision. Section VI discusses ways to combine data from multiple sources.

Results are shown on both simulated data and real data collected from the University of Canterbury's KiwiSAS III in Sydney and Lyttelton Harbours.

II. SAS BACKGROUND

The data for synthetic aperture sonar (SAS) images is collected from an acoustic transmitter and receiver pair or array mounted on a towfish, towed underwater behind a boat. In this paper, we are assuming a sidescan, stripmap system. SAS reconstruction algorithms assume the towfish travels in a straight path at a constant velocity. Unknown motion of both the towfish and the imaging medium cause blurring of the images obtained. The major contributors to this blurring are unknown speed, as this affects along-track sampling, and unknown sideways motion, or sway, as this directly affects the phase of the returned signal. These two motions are discussed in this paper, but the method is easily extended to other motion parameters.

Incorrect sway or velocity estimates blurs the image in the along-track direction. Fig. 1(a) shows the effect of uncorrected towfish sway on the reconstruction of a point target. When the towfish sway is known, it is possible to invert its effect on an image by assuming a narrow beam and no slant-range variation [4] using

$$\hat{D}(f, y) = \tilde{D}(f, y) \exp[-j4\pi(f + f_0)\hat{w}(y)/c]. \quad (1)$$

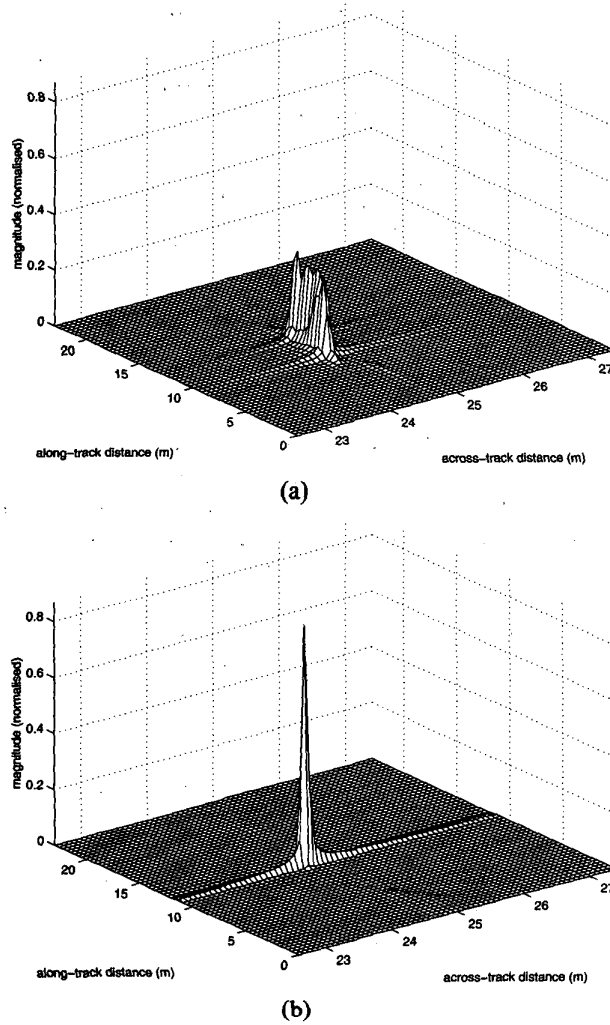


Fig. 1. Reconstructed image of a simulated single point target (a) with sway (max sway 0.14 m), (b) with known sway corrected.

Here $\tilde{D}(f, y)$ is the temporal Fourier transform of $\tilde{d}(t, y)$, the measured pulse-compressed baseband data, $\hat{w}(y)$ is the estimated sway as a function of along-track position y , f is the baseband frequency, f_0 is the centre frequency, and c is the speed of sound in water. Fig. 1(b) shows the blurred image in Fig. 1(a) corrected for the known sway, using (1).

III. STATISTICAL VIEWPOINT

Given the blurred, pulse-compressed data $\tilde{d}(t, y)$, the most likely estimate of sway, $\hat{w}(y)$, is desired. Using a statistical approach we seek,

$$\hat{w}_{MAP}(y) = \max_{\hat{w}} \left(\Pr\{\hat{w}(y) | \tilde{d}(t, y)\} \right). \quad (2)$$

Using Bayes' theorem, and taking logs, we obtain,

$$\hat{w}_{MAP}(y) = \max_{\hat{w}} \left(\log[\Pr\{\tilde{d}(t, y) | \hat{w}(y)\}] + \log[\Pr\{\hat{w}(y)\}] \right). \quad (3)$$

This is a MAP (maximum a posteriori) estimator, with $\Pr\{w(y)\}$ incorporating prior information on the path variation. It can be calculated from the statistical properties of the towfish. If we assume Gaussian statistics,

$$\log[\Pr\{\hat{w}(y)\}] = -\frac{1}{2} \hat{w}^T \mathbf{R}^{-1} \hat{w}, \quad (4)$$

where $\mathbf{R} = E[\mathbf{w}\mathbf{w}^T]$ is the covariance of the sway, and $\hat{w}$ is a column vector of the estimated sway at each along-track position y . This term acts as a form of regularisation by penalising large or rapid movements of the towfish. Fig. 3 shows the log-likelihood surface used, over the first two path parameters. The greater the movement from centre, the lower the path probability.

The log-likelihood of an image, given a sway estimate, also needs to be determined. The approach taken in this paper is first to form an image estimate, $ss(x, y)$. This is done by correcting the pulse-compressed image with the estimated sway using (1) and then reconstructing the image using a SAS algorithm such as the wavenumber algorithm [4,5], effectively an inverse model of the imaging system. It is then possible to make a measure of the quality of the reconstructed image to estimate the image's log-likelihood.

IV. MEASURING IMAGE QUALITY

With no prior knowledge of the composition of the image and in the context of varying focus parameters, a simple and effective way to estimate image quality is by measuring image contrast [3,6]. The following image contrast measures were considered. Meaningful quantitative comparisons are difficult, so some qualitative comparisons are noted gained by observation and experience.

1. The maximum value of the image amplitude.

$$C_1 = \max |ss(x, y)| \quad (5)$$

This metric is useful in astronomy applications, when a single target is being resolved, but is a poor measure when there is more than one target, as all but the brightest target are ignored.

2. Sum of image intensity.

$$C_2 = \frac{\iint |ss(x, y)|^2 dx dy}{\left(\iint |ss(x, y)| dx dy \right)^2} \quad (6)$$

This metric performed best in most trials.

3. The square of the image intensity.

$$C_2 = \frac{\iint |ss(x, y)|^4 dx dy}{\left(\iint |ss(x, y)| dx dy\right)^4} \quad (7)$$

Because $|ss(x, y)|^2$ is the intensity of the image, this metric is equivalent to the classic Muller-Buffington squared-intensity image sharpness metric [6]. However it was observed in a multiple target image, that this metric gives too much weight to the strongest target to the detriment of other targets in the image.

4. The negative of the entropy of the image [2].

$$C_4 = \iint I(x, y) \ln[I(x, y)] dx dy \quad (8)$$

Where the power normalised image $I(x, y)$ is

$$I(x, y) = \frac{|ss(x, y)|^2}{\iint |ss(x, y)| dx dy}.$$

The C_4 metric did perform satisfactorily but in general did not give as well defined a peak as C_2 .

5. The ratio of the standard deviation to the mean of the amplitude [3].

$$C_5 = \frac{\sqrt{\iint [|ss(x, y)| - \iint |ss(x, y)| dx dy]^2 dx dy}}{\iint |ss(x, y)| dx dy} \quad (9)$$

This metric performed in a similar way to C_2 with the cost of extra complexity.

V. OPTIMISATION ALGORITHM

The method used to solve the problem in (3) is shown in Fig. 2. The measured data is corrected with the current path estimate, and an image estimate is formed. An image-quality metric is measured from the image. This is combined with the cost of the path estimate, to obtain a cost measure using (10). This measure is fed into an optimisation routine programmed to minimise this cost measure by perturbing a set of parameters, representing the path estimate. The cost function is calculated using

$$\begin{aligned} \text{Cost} &= -(\log[\Pr\{\tilde{d}(t, y) | \hat{w}(y)\}] + \log[\Pr\{\hat{w}(y)\}]) \\ &\approx -(C_k + \eta \times \log[\Pr\{\hat{w}(y)\}]) \end{aligned} \quad (10)$$

Where C_k is an image contrast measure, $\log[\Pr\{\hat{w}(y)\}]$ is given by (4) and η is a constant balancing the effect of the two components. The optimisation algorithm needed is an unconstrained non-linear optimisation of several variables,

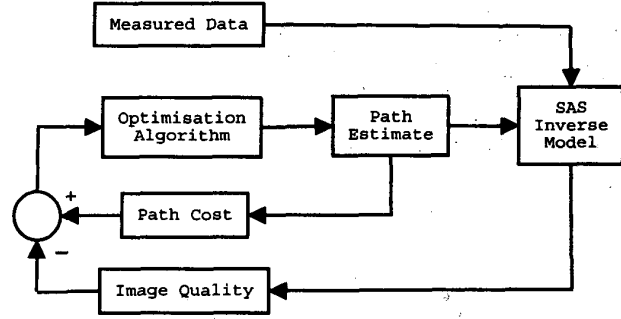


Fig. 2. Method to solve inverse problem using statistical optimisation algorithm (MAP estimator).

which is well dealt with in the literature. We used a standard multi-variable, quasi-Newton minimisation algorithm [7].

This autofocus technique is comparable to adaptive optics schemes used for real-time correction of atmospherically degraded telescope images through image sharpening [6]. However, the phase variations are introduced to the image in post-processing through an optimisation algorithm, rather than a real-time feedback system. An extra difficulty is that the blurring is point-spread-variant, meaning the optimising elements are highly coupled. It is different from other stripmap contrast optimisation schemes [8], in that it treats the path as a whole, rather than breaking it into strips, so it avoids tight restrictions on scatterer positions. Rather than represent the path by simple low-order polynomial terms [9], the algorithm can autofocus an arbitrary path up to the desired precision, by including enough path parameters in the optimisation. Due to the point-spread-variant nature of the blurring, we have been unable to calculate an analytic gradient. This rules out using a highly efficient conjugate-gradient optimisation algorithm [10,11].

Figures 6 and 8 show two typical contrast surfaces over the first two parameters. Fig. 6 is for a simulated image, which gives a smooth surface, with a single, well defined peak. This is found easily by an optimisation routine. Fig. 8 is for a real image. This gives a single, major peak, but the surface appears quite noisy. A local gradient-based optimisation routine will struggle to find the peak of this function.

A. Parameterising the Towfish Motion

Perturbing each individual position of the path is very inefficient as each point is highly correlated with the point next to it. It would be much faster if the path could be parameterised by fewer independent coefficients. This suggests representing the path as a series of basis functions and only perturbing those functions that have higher mean energy. However the type, and number of basis functions used, is dependent on the statistics of the towfish motion. For example, a fish that moves rapidly requires more terms to represent its path than one that moves smoothly.

The optimal set of basis functions for the path are the eigenvectors of the covariance matrix of the path, $\mathbf{R}$ [12]. The eigenvalues represent the mean-square energy of the corresponding term. By choosing the eigenvectors with this higher mean-square energy, you can represent the path accurately by the fewest terms. To choose enough parameters to represent 95% of the energy of the path, you choose the number of eigenvalues with 95% of the total.

The path $\hat{\mathbf{w}}$ can be represented by

$$\hat{\mathbf{w}} = \mathbf{U}\hat{\mathbf{b}}, \quad (11)$$

where $\mathbf{U}$ is assembled from columns of the eigenvectors of $\mathbf{R}$ and $\hat{\mathbf{b}}$ is a column vector of coefficients, obtained using, $\hat{\mathbf{b}} = \mathbf{U}^{-1}\hat{\mathbf{w}} = \mathbf{U}^T\hat{\mathbf{w}}$. This method is very flexible as any parameters of the inverse model may be used, for example towfish velocity, yaw, sway, pitch, heave, and roll.

B. Towfish Statistics

The log-likelihood of the path term acts as a regularisation term, penalising large or rapid movements in the cost function. Fig. 3 shows how the log-likelihood of the path varies over the first two path parameters. This is scaled by η and added to the contrast surface to give the negative of the cost function. The effect of the towfish statistics on the cost function, including the effect of altering η , is shown in Fig. 4.

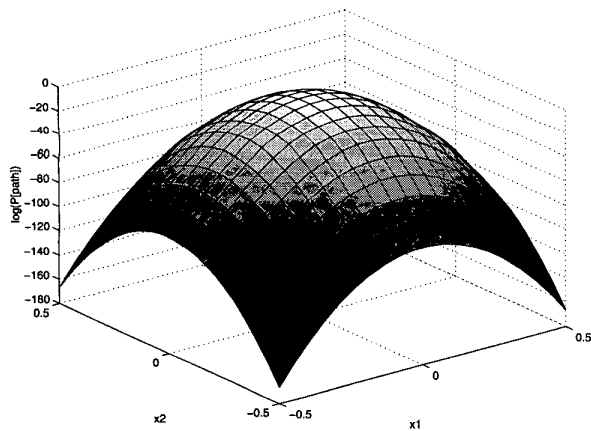
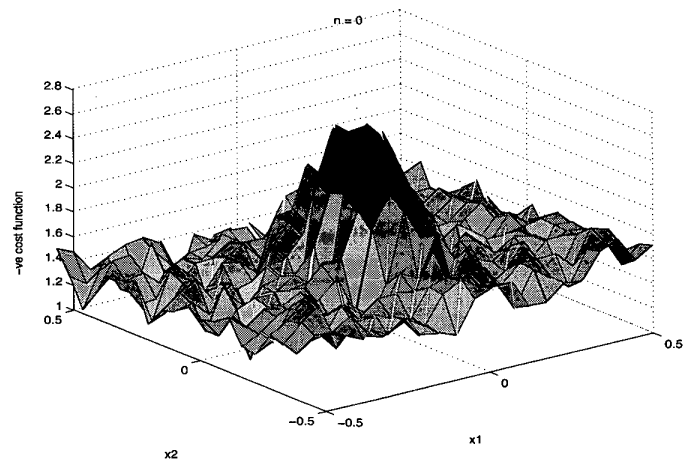
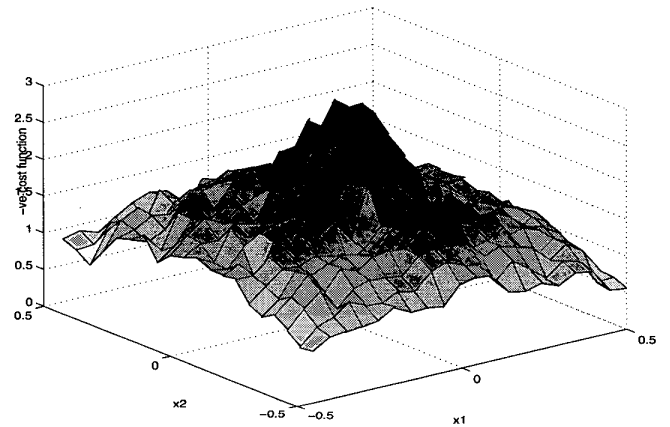


Fig. 3. Log-likelihood of path. x_1, x_2 are first two path parameters.

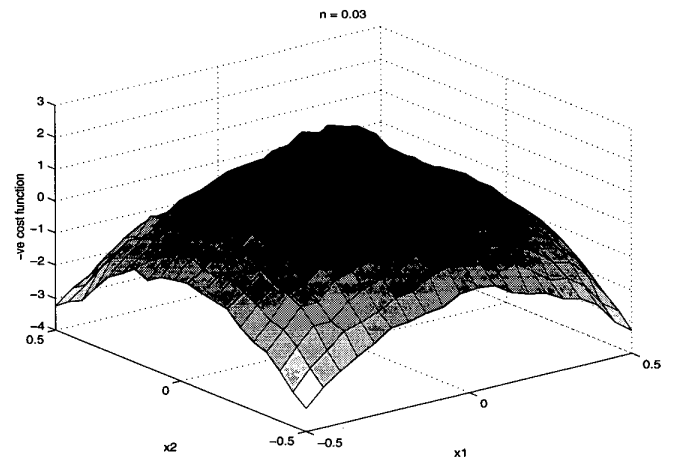
This is typical of real images. Outside of the central peak, the cost function is flat and noisy. Adding the regularisation term gives this flat section some gradient, allowing the optimisation algorithm to move the solution towards the peak. The solution was found to be insensitive to η up to a point, where over-regularisation occurs and the path cost term dominates the solution.



(a) $\eta = 0.007$



(b)



(c)

Fig. 4. Negative cost function for real image over first two path parameters.

(a) $\eta=0$ (no regularisation). (b) $\eta=0.007$ (small regularisation).

(c) $\eta=0.03$ (over regularisation).

C. Using Bulk Error Removal

The contrast function of real images tends to have a confined domain of attraction. This can be seen in Figures 4, 5 and 8. If the initial estimate for the optimisation algorithm fell outside this region, it would not converge to the peak. If it did, it would converge very slowly due to the small gradient far from the peak. It is desirable for the initial estimate to be quite close the peak. To achieve this, we propose the use of inaccurate inertial navigation system (INS) measurements, or a fast (but less accurate) bulk-error removal algorithm, such as noncoherent shear-average [13,14], to give an initial estimate in the optimisation algorithm. Fig. 7 shows an example of using this fast shear-average algorithm as an initial estimate to the contrast optimisation algorithm.

VI. MULTIPLE FREQUENCY, INTERFEROMETRIC SYSTEM

Few modern SASs consist of a single transmitter and receiver system, mainly due to mapping rate restrictions. The KiwiSASIII system, used by the authors, images in two separate bands; 30kHz and 100kHz. It does this from three vertically separated receivers to collect interferometric information. This system gives six separate channels, for six separate looks at the image. Sway and velocity errors should affect each channel equally, giving six separate estimates of sway.

There are three ways the data can be combined into one estimate. Each channel could be autofocussed separately, then the different paths combined into a best estimate. Fig. 5(a) shows the contrast surface for a single channel. The surface is very noisy, and will be difficult to optimise. Another approach would be for a given sway estimate, all channels are reconstructed and the contrast measures then combined. The advantage of this approach is that noise, and local gradient errors are reduced. However if the channels have peaks that are distinct, the result may have multiple peaks. Fig. 5(b) shows the sum of the contrast surface of all the channels. It is observed that the sum of the channels gives a smoother surface, but the peak is wider and is bimodal. A third approach would be to optimise the contrast of the noncoherent sum of the different images. This is shown in Fig. 5(c). These are only preliminary results, and the best method has not been determined. It appears that summing the contrasts of the images gives the smoothest surface, which is the easiest to optimise, but optimising each surface will probably give a more accurate result if it is possible.

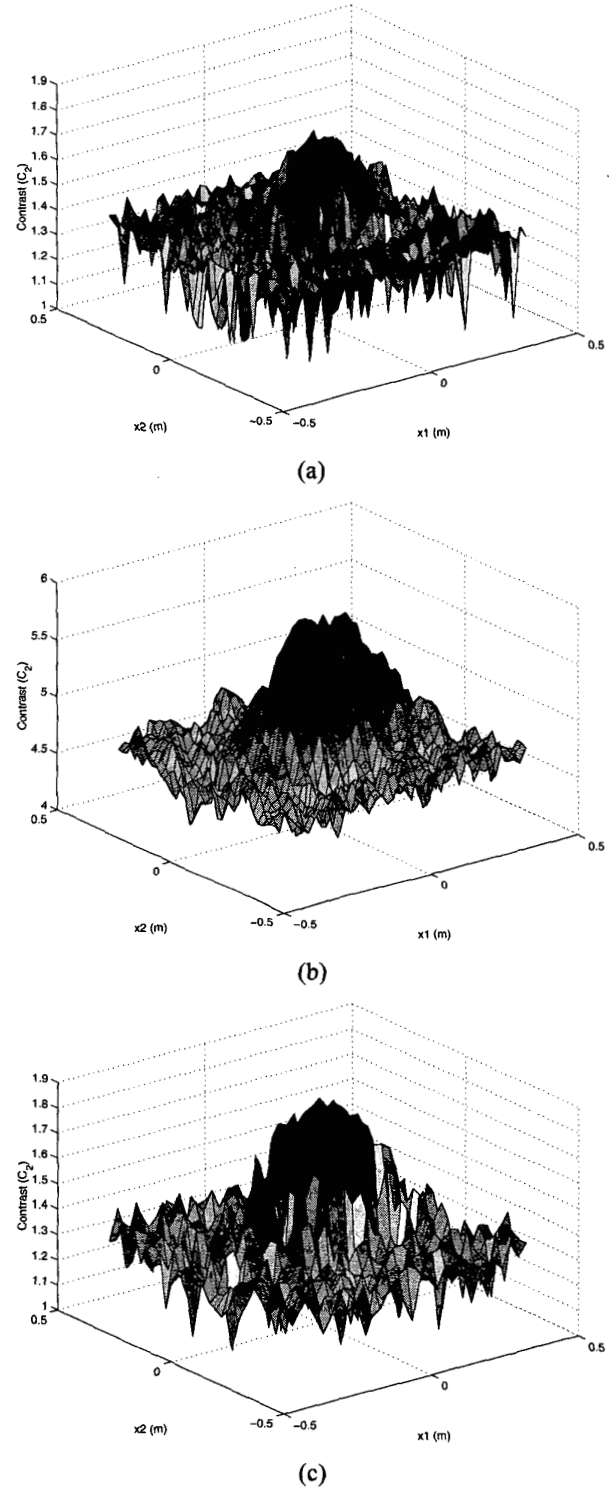


Fig. 5. Contrast Surface over first two path parameters of real multiple-frequency, interferometric image. (a) Single channel (30kHz top receiver). (b) Sum of contrast of all channels. (c) Contrast of noncoherent sum of channels.

VII. RESULTS

A simulated image was created with 50000 point-targets modelling seabed clutter, with four strong targets superimposed. The towfish path had a known, randomly created sway. Fig. 7(a) shows the simulated image with sway. Figures 7(b) and 7(c) show the image with the sway compensated using an initial estimate, and the final estimate of the statistical autofocus algorithm. This matches the actual sway very closely, giving an image imperceptible from ideal. Fig. 6 shows the contrast function mapped over the first two path parameters. This shows a smooth surface, with a distinct peak. The actual sway parameters do not lie on the peak, as all other path parameters are held at zero for this plot. The estimate was obtained using 15 parameters, and hence matched the solution a lot closer.

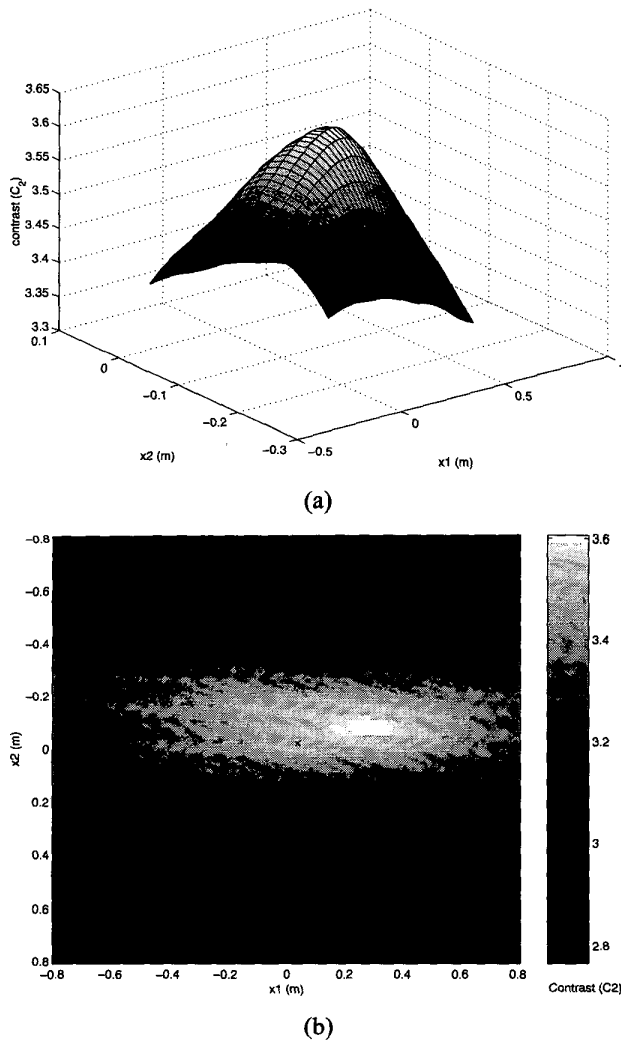


Fig 6. Contrast Surface over first two path parameters of simulated image. (a) Surface Plot. (b) Grayscale image. 'x' marks the actual sway.

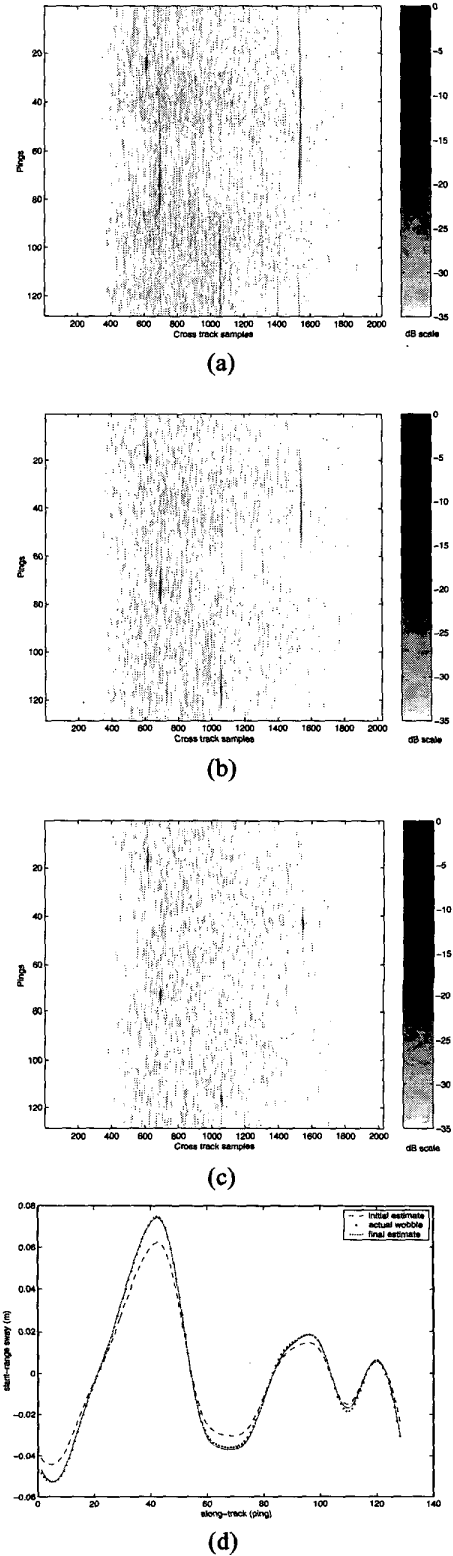


Fig. 7. Simulated image with sway. (a) Original image reconstructed. (b) Image reconstructed with initial shear-average estimate. (c) Image reconstructed with final estimate. (d) Actual and estimated paths.

The next result shown is from a real image. This particular image was one with very little sway, which gave a very good quality result. Known sway of only two parameters was then introduced into the image. The contrast was then mapped as these two parameters were varied. Fig. 8 shows the resulting contrast surface. The peak of the contrast matches very closely to the introduced sway, (within the 1cm resolution mapped). However, due to the spikey, noisy nature of the surface, the optimisation algorithm was unable to find the peak and stopped at a local maximum close to the initial estimate of zero.

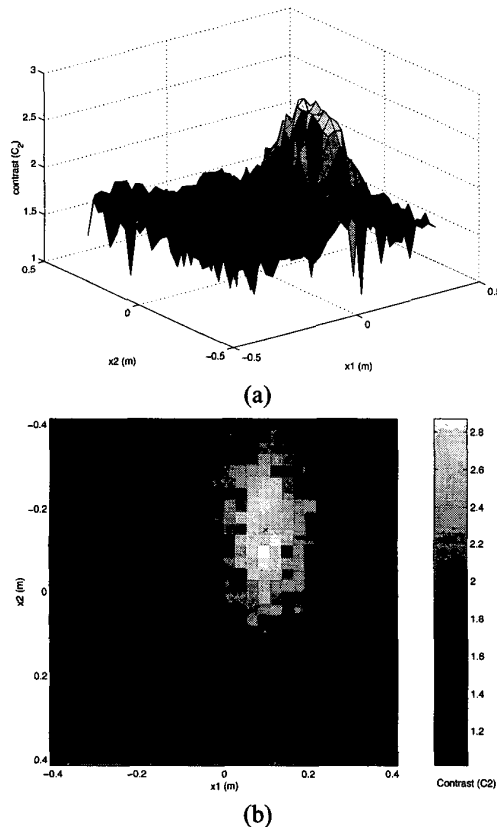


Fig. 8. Contrast variation over first two path parameters of real image, that has had sway introduced. (a) Surface plot. (b) Grayscale image. 'x' marks the actual sway introduced into the image.

VIII. CONCLUSIONS

The technique of image-contrast optimisation has been presented in a general Bayesian framework. This varies from similar autofocus schemes in that it includes the statistics of the towfish motion with the addition of a regularisation term (the log-likelihood of the path estimate). The path is also optimally represented by the fewest parameters using the towfish motion statistics.

Results showing the image contrast surface for real data demonstrates the need for this regularisation. It also demonstrates the need for some bulk-error removal method to

initialise the algorithm. This is easily done by a noncoherent shear-average technique [14], or INS measurements.

Several image contrast measures have been tried, and the best has been found to be the normalised average image intensity.

Results on simulated images, show a smooth, well-behaved cost-function, which is easily optimised using standard techniques. This gives a very accurate, although slow autofocus technique. Results on real images show a very noisy cost-function. Local gradient optimisation algorithms fail to minimise these functions. A peak exists however, so the technique of contrast optimisation is still viable. A global optimisation algorithm, rather than a local one, will need to be employed.

Preliminary results were shown on combining the data from a multiple-channel system. This was shown to have the effect of lowering the noise on the cost-function.

The algorithm is quite computationally expensive, so lends itself to use for obtaining high-quality imagery through post-processing of regions of interest.

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